

● EXECUTIVE REPORT · DATA ANALYTICS

EXECUTIVE BUSINESS INSIGHTS REPORT

Stock Performance Analysis

An End-to-End Data Analytics Project — presenting analytical findings, dashboard insights, and business recommendations derived from historical market data.

PREPARED BY

Lohith

PROJECT DURATION

June 2026 – July 2026

PYTHON

MYSQL

SQL

POWER BI

GIT & GITHUB

STOCK PERFORMANCE DASHBOARD

Confidential | Executive Insights Review

Table of Contents

■ 01	Executive Summary	3
■ 02	Project Overview	4
■ 03	Dashboard Overview	5
■ 04	Data Analysis Using Python	7
■ 05	Business Insights	12
■ 06	Dashboard Insights	13
■ 07	Business Recommendations	14
■ 08	Future Scope	15
■ 09	Conclusion	16
■ A	Appendix — Project Workflow	17



SECTION 01

Executive Summary

Introduction

This report summarizes the findings of the Stock Performance Analysis project, an end-to-end data analytics solution designed to evaluate the historical performance of five leading technology companies over a one-year period.

The project demonstrates the complete analytics lifecycle by integrating automated data collection, relational database storage, KPI generation, statistical analysis, and interactive business intelligence reporting.

Historical market data was collected using Python, stored within a MySQL database, analyzed using custom analytical scripts, and visualized through Microsoft Power BI dashboards to generate meaningful business insights.

The objective of this report is to present the analytical findings and demonstrate how data-driven visualization can support better investment analysis and business decision-making.

5COMPANIES
ANALYZED**1 YR**

ANALYSIS PERIOD

2POWER BI
DASHBOARDS**5**KEY BUSINESS
INSIGHTS



SECTION 02

Project Overview

The project analyzed historical stock performance for five major technology companies.

Company	Stock Symbol
Apple	AAPL
Microsoft	MSFT
Alphabet (Google)	GOOGL
Amazon	AMZN
Tesla	TSLA

Dataset Characteristics

- Historical daily prices
- Open price
- Closing price
- High price
- Low price
- Trading volume
- Daily returns
- Moving average
- Volatility
- Total returns

One Year

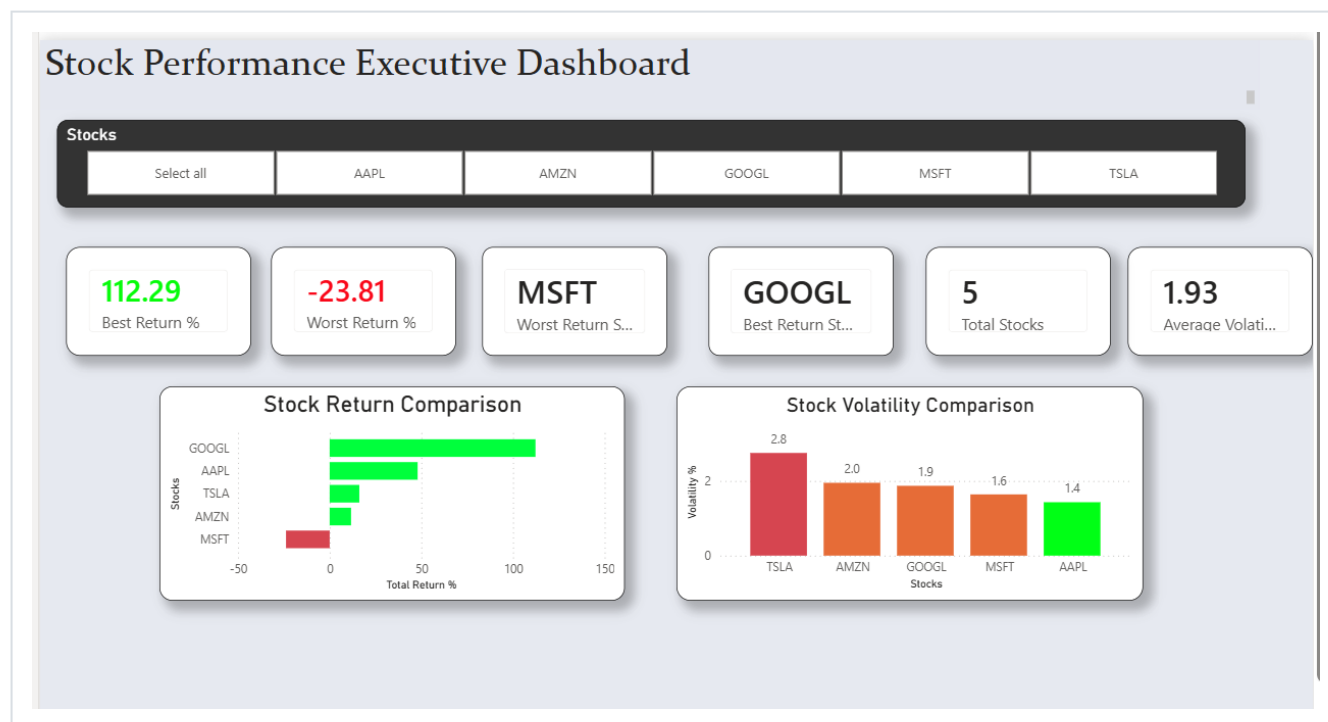
ANALYSIS PERIOD



SECTION 03

Dashboard Overview

Executive Dashboard

**FIGURE 1** Executive Dashboard

The Executive Dashboard provides a high-level overview of stock market performance through interactive KPI cards, comparative visualizations, and dynamic filtering.

The dashboard enables business users to:

- Compare stock performance
- Monitor historical trends
- Evaluate market volatility
- Analyze trading activity
- Filter results by stock symbol

Trend Dashboard



FIGURE 2 Trend Analysis Dashboard

The Trend Dashboard focuses on historical movement across the selected companies by displaying:

- Closing price trend
- Trading volume trend
- Interactive stock filters

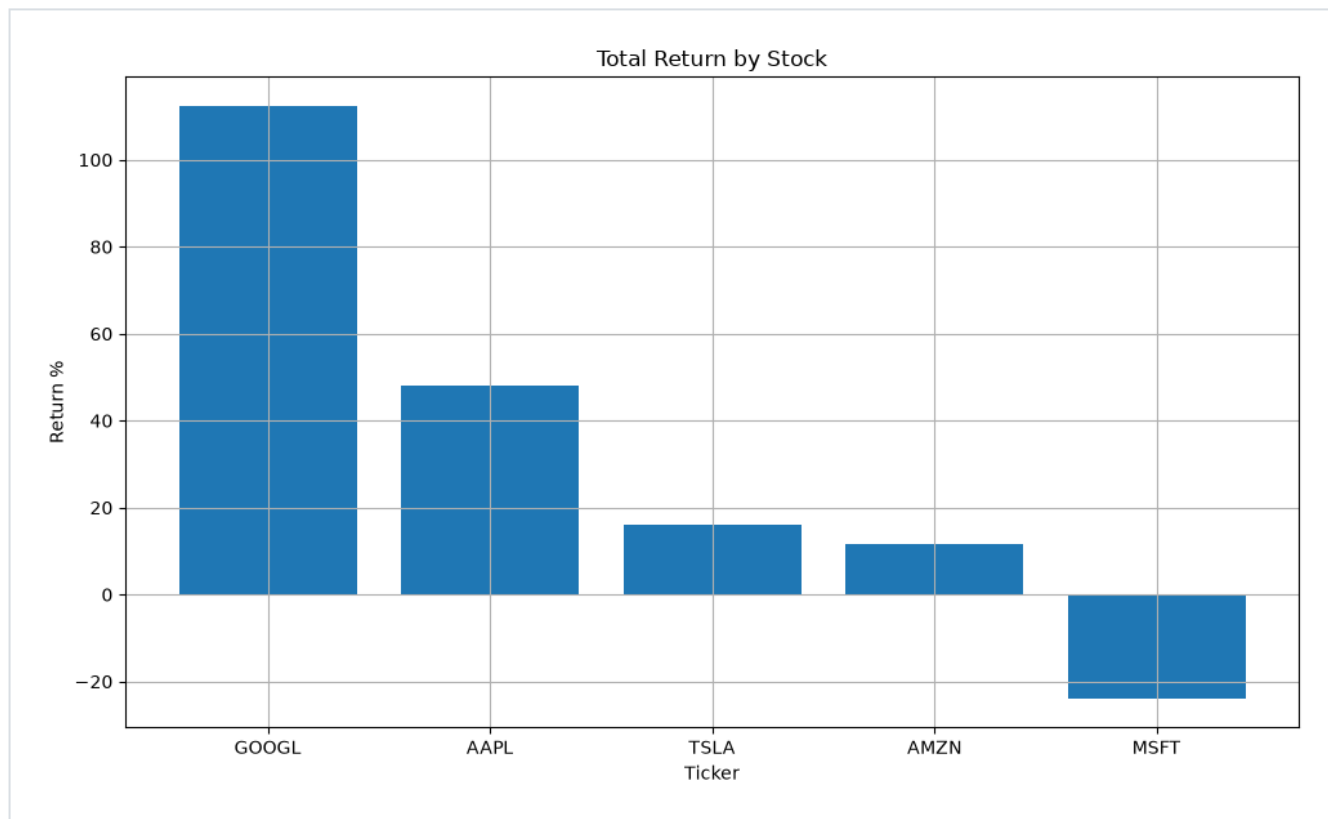
This enables users to identify long-term price behavior and market activity.



SECTION 04

Data Analysis Using Python

4.1 Total Return Comparison

**FIGURE 3** Total Return Comparison

The Total Return comparison measures the overall percentage growth achieved by each company during the selected analysis period.

KEY FINDINGS

- Alphabet (GOOGL) delivered the highest overall return.
- Apple generated strong positive growth.
- Tesla maintained moderate growth.
- Amazon remained relatively stable.
- Microsoft produced the lowest return.

4.2 Volatility Comparison

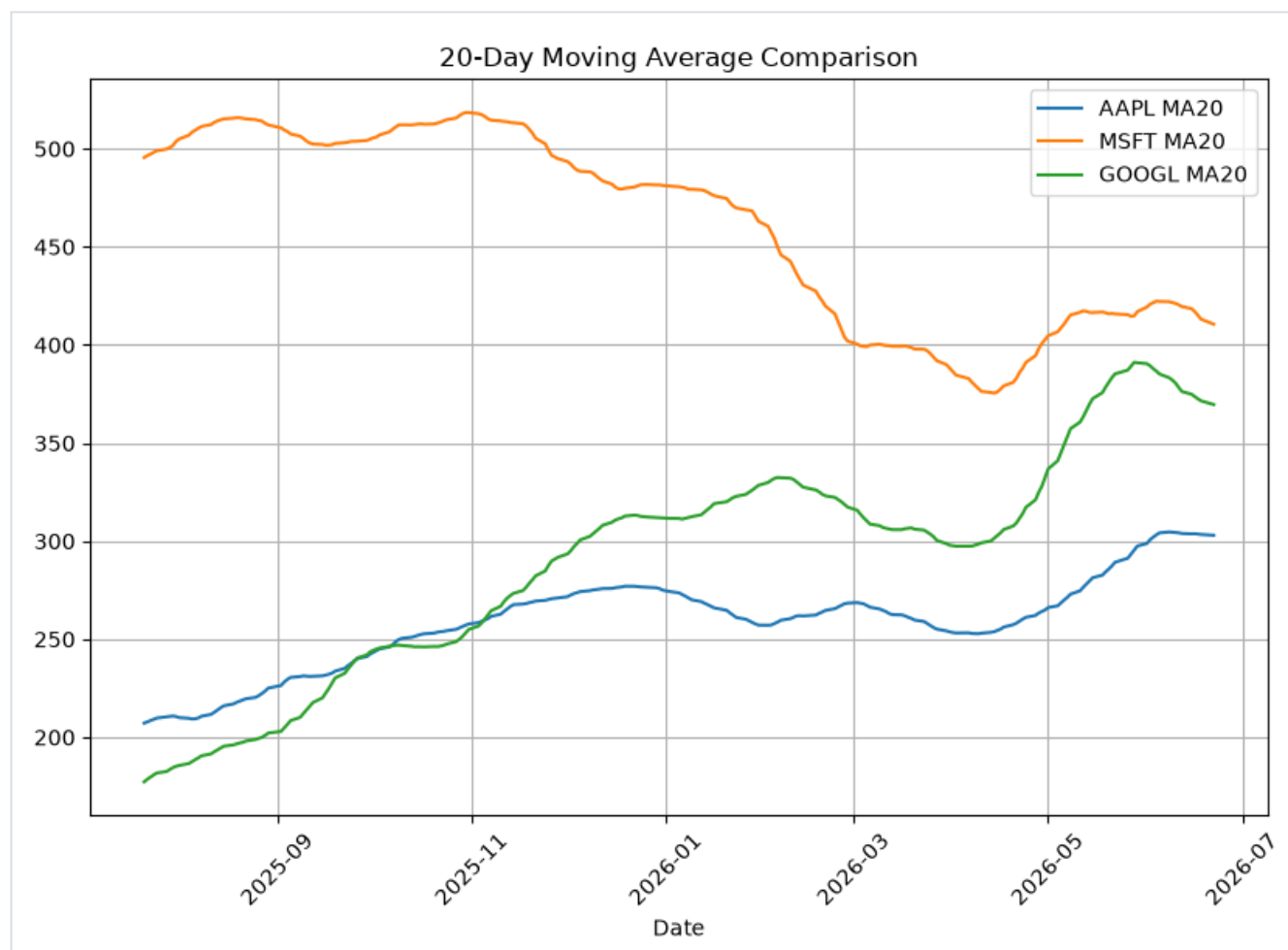


FIGURE 4 Stock Volatility Comparison

Volatility measures the degree of price fluctuation experienced by each stock. Higher volatility generally indicates higher investment risk.

KEY FINDINGS

- Tesla recorded the highest volatility.
- Apple exhibited the most stable price movement.
- Microsoft and Amazon demonstrated moderate fluctuations.

4.3 Trading Volume Comparison

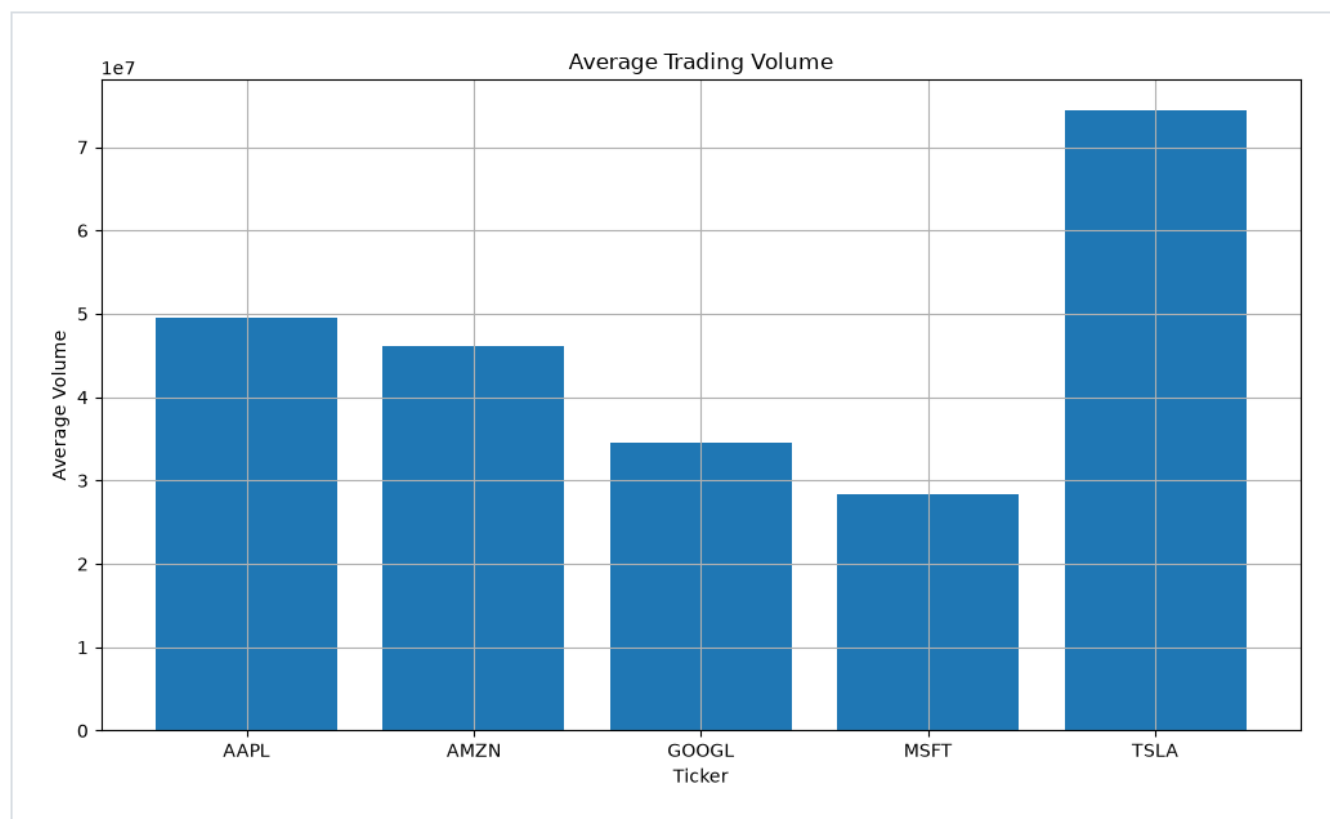


FIGURE 5 Average Trading Volume

Trading volume reflects investor participation and market liquidity.

KEY FINDINGS

- Tesla consistently recorded the highest trading volume.
- Apple also demonstrated significant trading activity.
- Microsoft maintained comparatively lower trading volume.

4.4 Moving Average Comparison

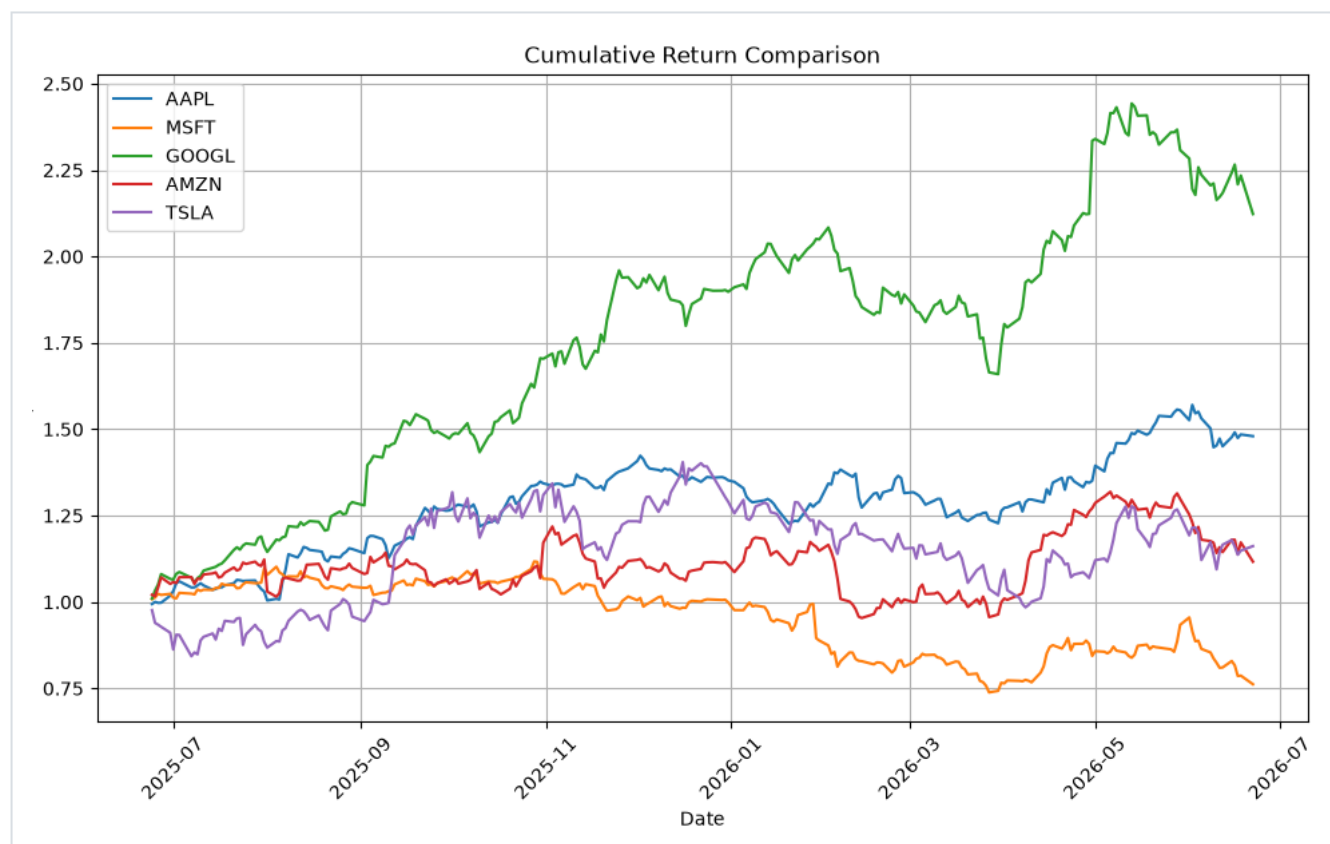


FIGURE 6 Moving Average Trend

Moving averages smooth short-term market fluctuations and highlight long-term price direction.

KEY FINDINGS

- Alphabet demonstrated the strongest upward trend.
- Apple maintained consistent long-term growth.
- Tesla showed larger fluctuations despite an upward movement.

4.5 Cumulative Return Analysis

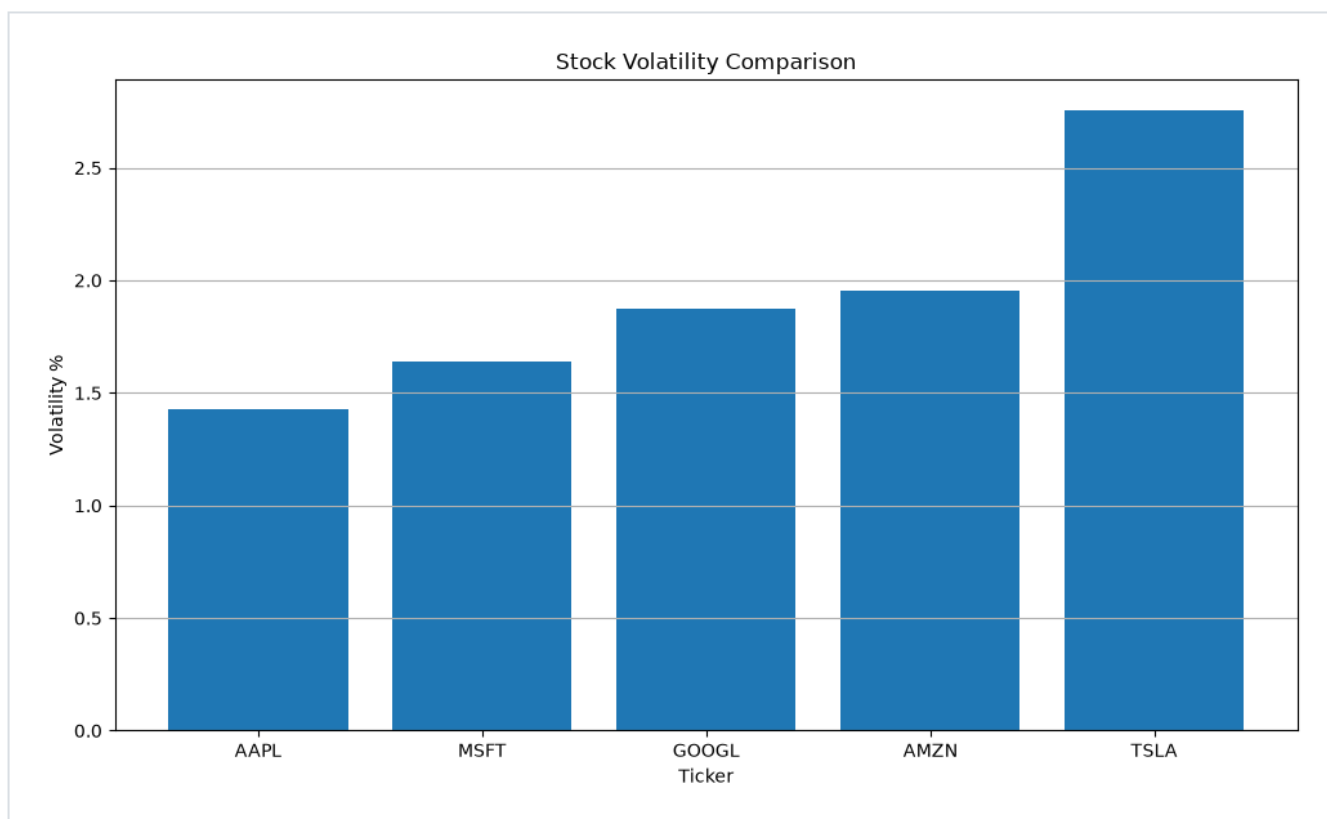


FIGURE 7 Cumulative Return Analysis

Cumulative Return illustrates how a hypothetical investment would have grown over the analysis period.

KEY FINDINGS

- Alphabet produced the greatest investment growth.
- Apple generated consistent long-term appreciation.
- Microsoft experienced periods of negative performance.



SECTION 05

Business Insights

The analytical results provide several meaningful business observations.

01**Alphabet Leads on Return**

Alphabet significantly outperformed the remaining companies in terms of total return.

02**Tesla Carries the Highest Risk**

Tesla exhibited the highest volatility, indicating increased investment risk despite positive returns.

03**Apple Offers Consistency**

Apple maintained relatively stable performance while generating steady growth, making it one of the more consistent performers.

04**Microsoft Underperforms**

Microsoft delivered the weakest overall return during the selected period.

05**Tesla Draws the Most Investor Interest**

Tesla consistently attracted the highest trading volume, reflecting strong investor interest and market activity.



SECTION 06

Dashboard Insights

The Power BI dashboard transformed raw financial data into interactive business intelligence by providing:

- | | |
|--------------------------|-----------------------------|
| ● Executive KPI Cards | ● Comparative Bar Charts |
| ● Trend Analysis | ● Interactive Slicers |
| ● Multi-stock Comparison | ● Time-Series Visualization |

The dashboard enables business users to analyze historical performance without manually processing raw datasets.



SECTION 07

Business Recommendations

Based on the analysis, the following recommendations are proposed:

- 1 Monitor long-term return trends rather than short-term market fluctuations.
- 2 Evaluate volatility alongside returns when assessing investment opportunities.
- 3 Expand the solution to include additional sectors and market indices.
- 4 Automate dashboard refresh through direct MySQL connectivity.
- 5 Incorporate technical indicators for enhanced financial analysis.



SECTION 08

Future Scope

Future enhancements may include:

Live API Integration	Automated ETL Scheduling
Direct Power BI-MySQL Connectivity	Predictive Analytics
Machine Learning-Based Forecasting	Technical Indicators (RSI, MACD, Bollinger Bands)
Portfolio Optimization Dashboard	Cloud Deployment



SECTION 09

Conclusion

The Stock Performance Analysis project successfully demonstrates a complete end-to-end analytics workflow by integrating Python, SQL, MySQL, and Power BI into a unified business intelligence solution.

The project automates historical data collection, performs analytical calculations, validates data quality, and delivers actionable insights through interactive dashboards.

Beyond technical implementation, the solution highlights the practical application of data analytics in supporting financial decision-making and investment performance evaluation.

The modular design also provides a scalable foundation for future enhancements such as live market integration, predictive analytics, and enterprise reporting.



APPENDIX

Project Workflow

The end-to-end pipeline that transforms raw market data into an executive-ready business insights report:

- 1 Yahoo Finance API**
Source of historical daily stock market data.
- 2 Python ETL**
Extraction, cleaning, and transformation of raw market data.
- 3 MySQL Database**
Centralized relational storage for structured financial data.
- 4 SQL Validation**
Checks for missing values, duplicates, and data consistency.
- 5 Python KPI Analysis**
Calculation of returns, volatility, moving averages, and volume metrics.
- 6 Power BI Dashboard**
Interactive visualization of KPIs and comparative performance.
- 7 Business Insights Report**
Consolidated executive findings and recommendations.

— END OF DOCUMENT —